Research Computing: Practical session 1

1 Basic Linux usage

In the following, I assume you have a standard Bash-shell terminal open. If you use a native Linux (e.g. Ubuntu) installation, or Windows Subsystem for Linux (WSL) this is fine. Mac OS X users may also be OK, but should not be surprised if some commands work differently.

You may need to refer to the `man` pages to find the correct options to some commands.

Try the following exercises from a Linux command-prompt.

1. Find your identity: username, user-id on the current machine.
2. List your home-directory.
3. What is the full path of your home directory?
4. List *all* the files in your home directory (there are some that are hidden by default).
5. Who owns the files in your home directory?
6. Who owns the files in `/usr/bin`?
7. What processes are running on your computer?
8. In your home directory, create a new folder called `Linux_exercises`.

```
apt install gcc;
```

9. Copy the file `/usr/bin/gcc` to that directory.
10. Move the file `/usr/bin/gcc` to that directory (then put it back afterwards if necessary).
11. What kind of file is `/usr/bin/gcc`?
12. What kind of file is `Linux_exercises/gcc`?
13. Is this a problem? How can you solve it?
14. Find all files larger than 1MB in your home directory and in sub-directories.

2 Text/data processing

1. Using `ls` and `head`, find the first 20 files (in alphabetical order) in `/usr/bin`.
2. Find the 10 most recently updated files in `/var/log`.
3. Use `tr` to remove all consonants from "Celine Dion" without typing `D` or `d`.

4. The file `/dev/random` contains an infinite number of random characters. Use `head` and `tr` to extract 20 alphanumeric characters to use as a hard-to-guess password. (If you corrupt your terminal, you can either close it or use `reset`.)

   [Note that this question may not work well on Mac OS X.]

5. Download the `Practical_1_data.tar.bz2` from Moodle and un-tar it with `tar -jxf Practical_1_data.tar.bz2`
6. Reproduce the example from the slides using `join_1.txt` and `join_2.txt`
7. Use the `paste` command to combine the contents of `message_1.txt` and `message_2.txt`
8. Look up the `banner` command; how did I produce the text files in the previous question?
9. The file `hello.C` needs some changes making to it as given in `change.diff`. Use `patch` to apply them.
10. Use `awk` on the file `find_some_lines.txt` to extract the third word from every line that starts with a word ending in 'e' and the second word from every line that starts with a word ending in 'g'.